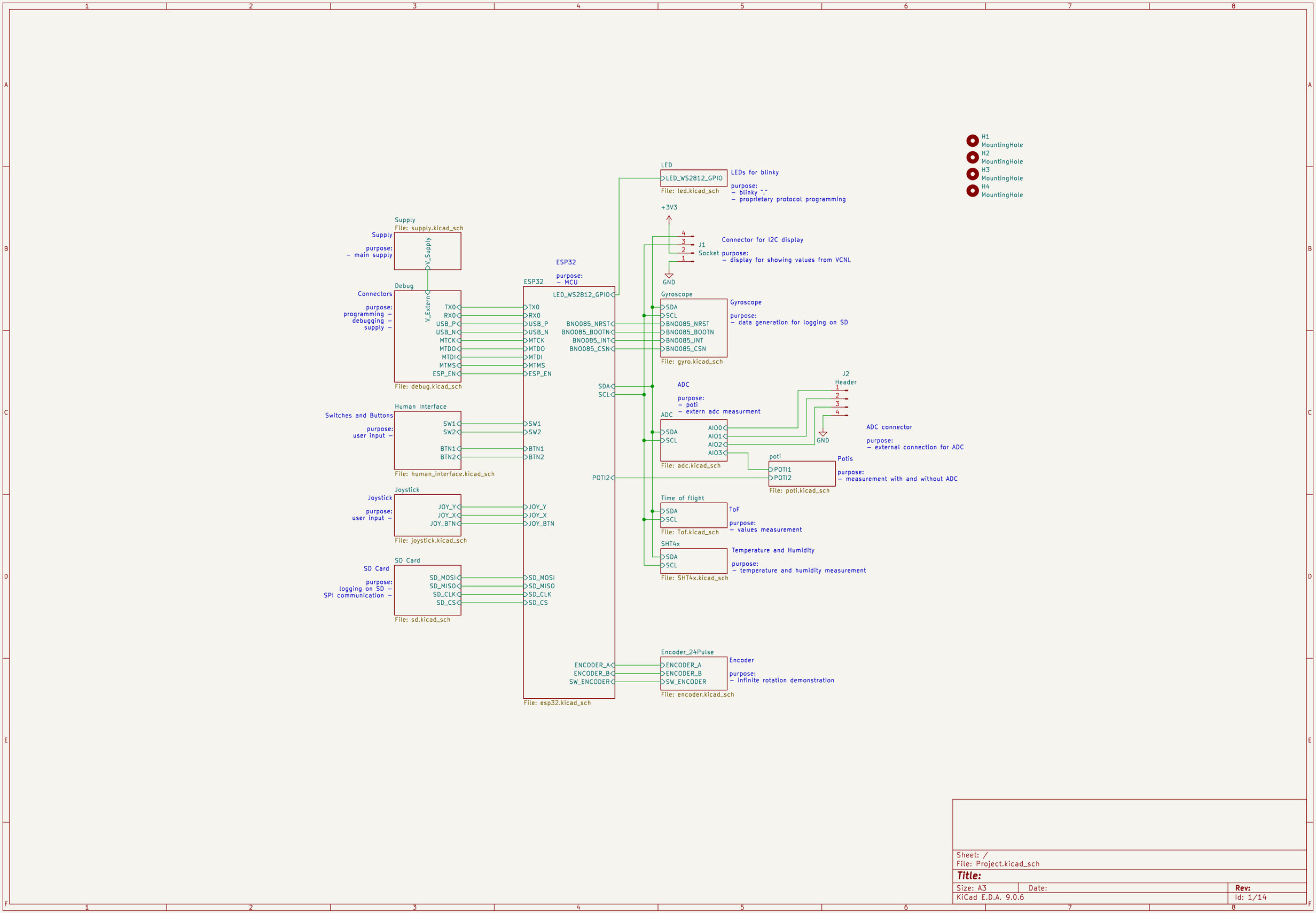
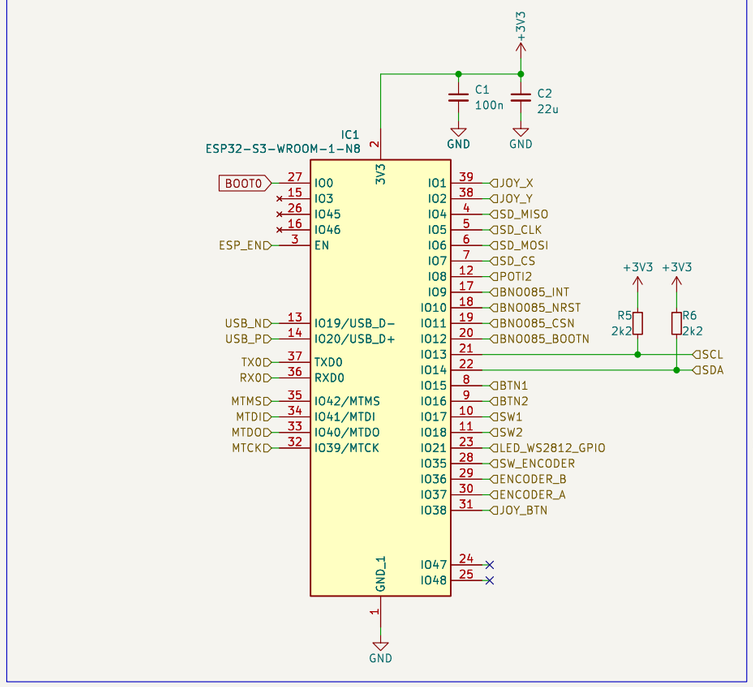
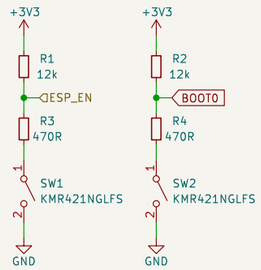

H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
LED
LED_WS2812_GPIO
LEDs for blinky
File: led.kicad_sch
purpose:
– blinky ^.^
– proprietary protocol programming
+3V3
4
3
2
1
J1
Connector for I2C display
Socket purpose:
– display for showing values from VCNL
GND
Supply
File: supply.kicad_sch
Supply
purpose:
– main supply
V_Supply
ESP32
purpose:
– MCU
ESP32
LED_WS2812_GPIO
Gyroscope
SDA
SCL
BNO085_NRST
BNO085_BOOTN
BNO085_INT
BNO085_CSN
Gyroscope
purpose:
– data generation for logging on SD
File: gyro.kicad_sch
Debug
Connectors
purpose:
programming –
debugging –
supply –
V_Extern
TX0
RX0
USB_P
USB_N
MTCK
MTDO
MTDI
MTMS
ESP_EN
File: debug.kicad_sch
J2
Header
1
2
3
4
ADC
purpose:
– poti
– extern adc measurment
SDA
SCL
ADC
AIO0
AIO1
AIO2
AIO3
GND
ADC connector
purpose:
– external connection for ADC
File: adc.kicad_sch
Human Interface
Switches and Buttons
purpose:
user input –
SW1
SW2
BTN1
BTN2
File: human_interface.kicad_sch
poti
Potis
POTI1
POTI2
purpose:
– measurement with and without ADC
File: poti.kicad_sch
Joystick
Joystick
purpose:
user input –
JOY_Y
JOY_X
JOY_BTN
File: joystick.kicad_sch
Time of flight
SDA
SCL
ToF
purpose:
– values measurement
File: Tof.kicad_sch
SHT4x
SDA
SCL
Temperature and Humidity
purpose:
– temperature and humidity measurement
File: SHT4x.kicad_sch
SD Card
SD Card
purpose:
logging on SD –
SPI communication –
SD_MOSI
SD_MISO
SD_CLK
SD_CS
File: sd.kicad_sch
Encoder_24Pulse
ENCODER_A
ENCODER_B
SW_ENCODER
Encoder
purpose:
– infinite rotation demonstration
File: encoder.kicad_sch
File: esp32.kicad_sch
Sheet: /
File: Project.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 1/14

ESP32-S3-WROOM-1U



Bootstrap

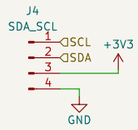


boot pin states are ONLY neccessary at system reboot.

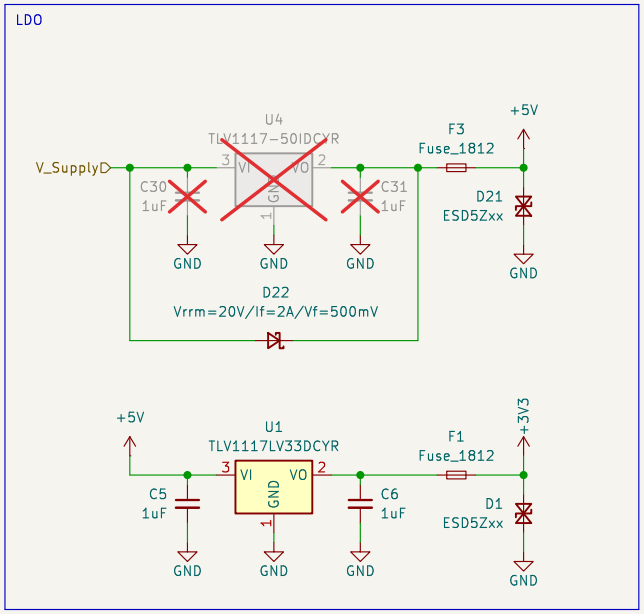
afterwards pins can be used normally after SPI boot in code.

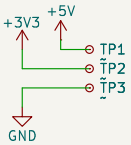
however mind the possible alteration of signals through the pull-up/-down resistors.

SDA&SCL Header

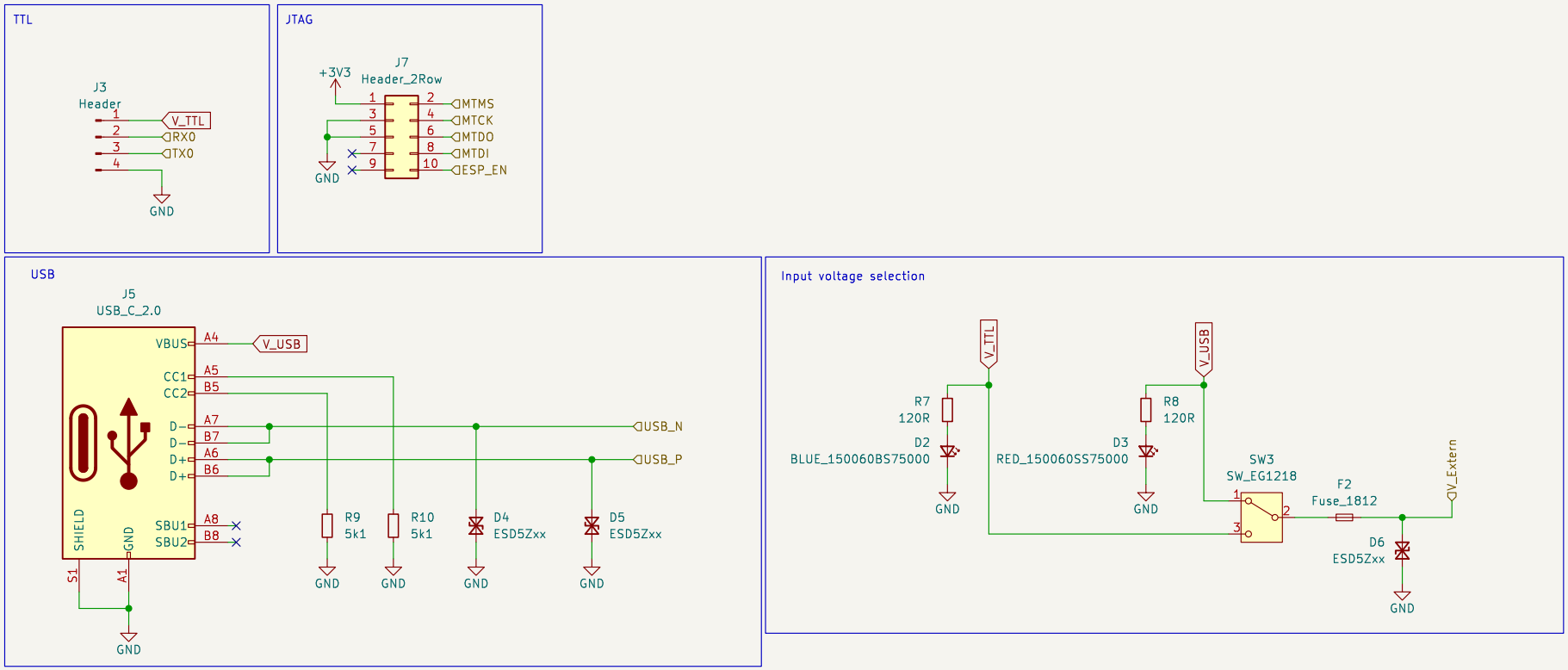


Sheet: /ESP32/		
File: esp32.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 2/14

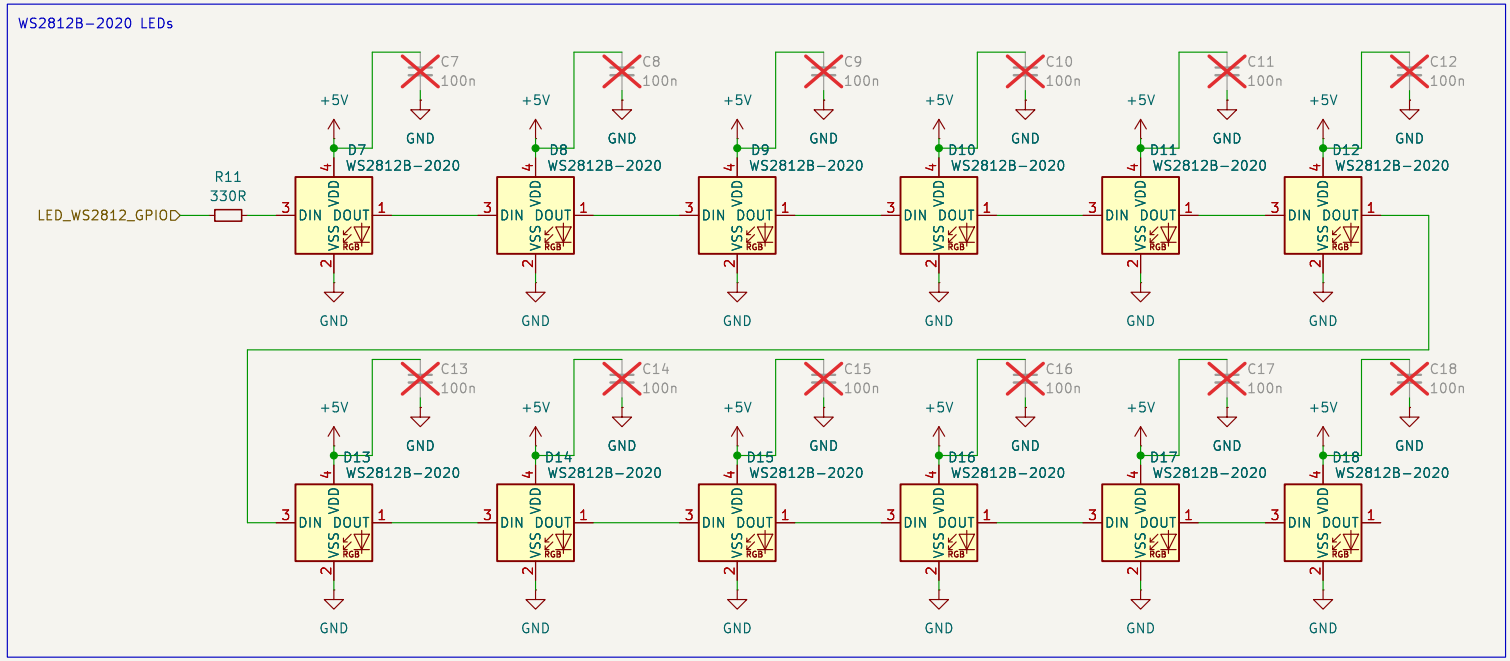




Sheet: /Supply/		
File: supply.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 3/14



Sheet: /Debug/		
File: debug.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 4/14



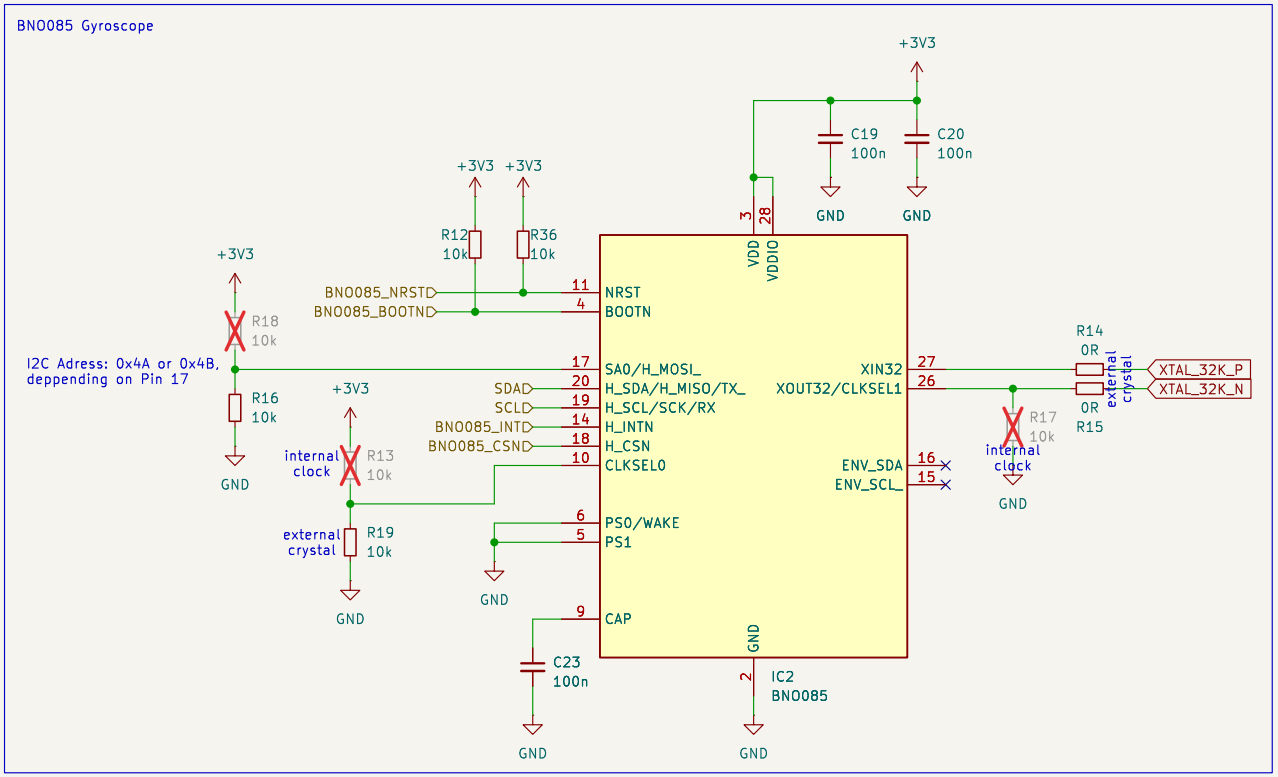
Sheet: /LED/
File: led.kicad_sch

Title:

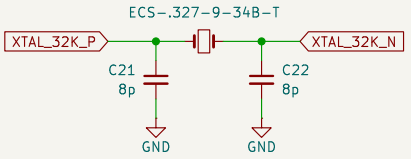
Size: A4 | Date: | **Rev:**

KiCad E.D.A. 9.0.6 | Id: 5/14

BNO085 Gyroscope



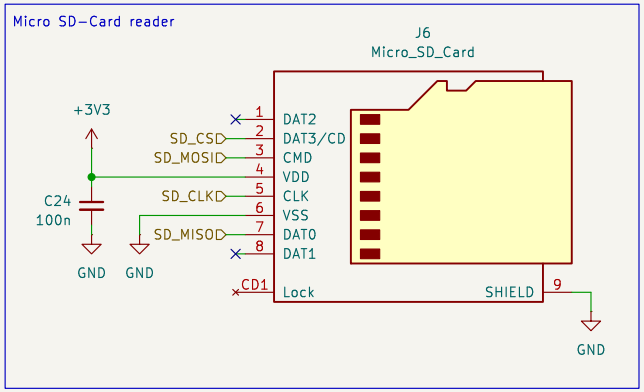
32.768 kHz Crystal



Sheet: /Gyroscope/
File: gyro.kicad_sch

Title:

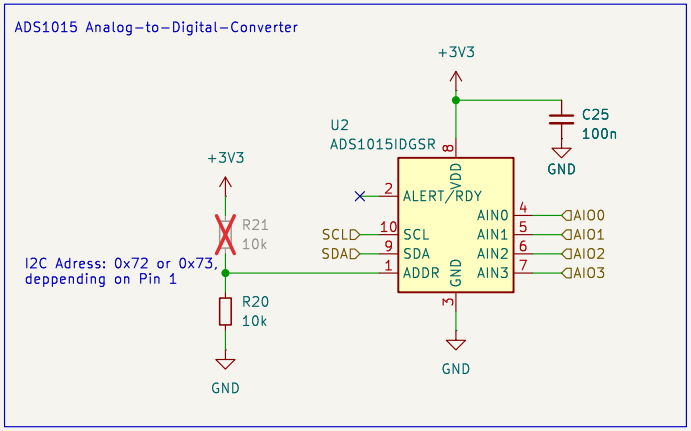
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 7/14



Sheet: /SD Card/
File: sd.kicad_sch

Title:

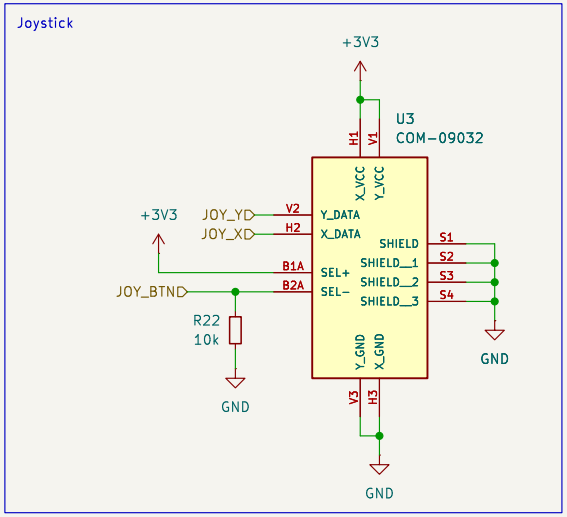
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 8/14



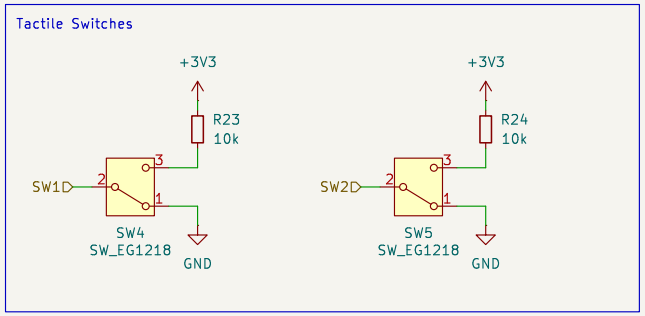
Sheet: /ADC/
File: adc.kicad_sch

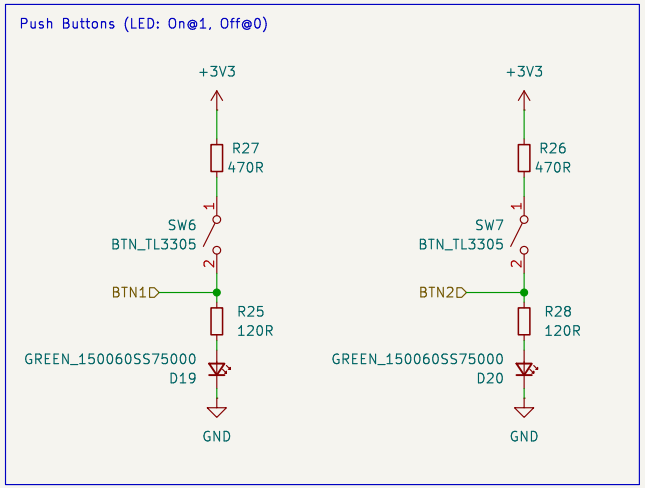
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 8/14



Sheet: /Joystick/		
File: joystick.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 9/14

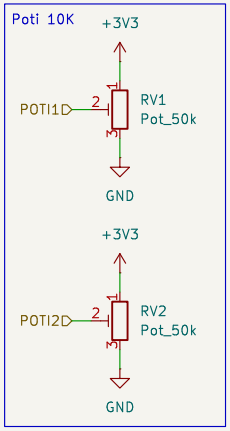




Sheet: /Human Interface/
File: human_interface.kicad_sch

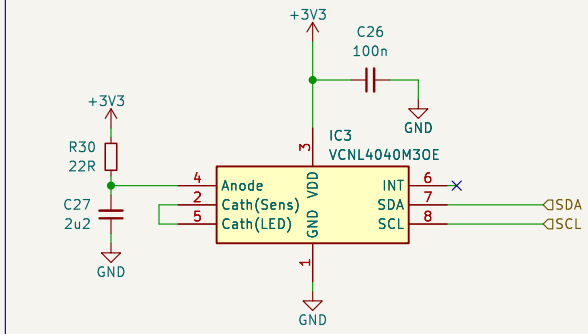
Title:

Size: A4	Date:	**Rev:**
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Sheet: /poti/		
File: poti.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 11/14

Time-of-Flight sensor with 20cm measurement distance



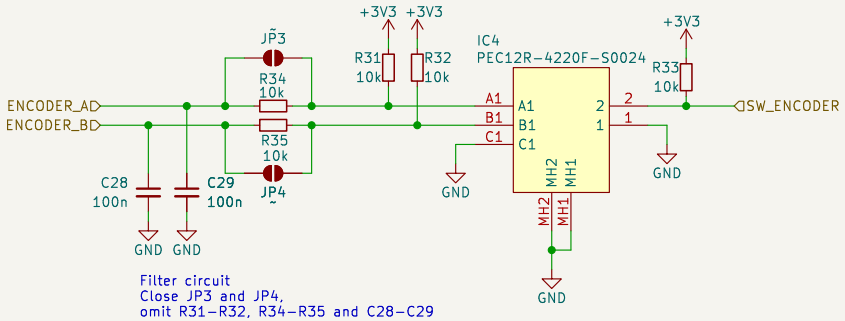
Cathode (LED) and cathode (sensor):
pins need to be connected together externally

Sheet: /Time of flight/
File: Tof.kicad_sch

Title:

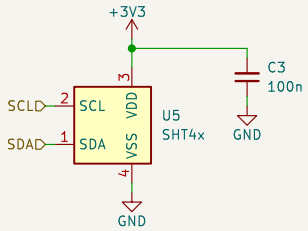
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 12/14

Encoder with 24 positions



Filter circuit
Close JP3 and JP4,
omit R31-R32, R34-R35 and C28-C29
to completely bypass the filter circuit.

Sheet: /Encoder_24Pulse/		
File: encoder.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 13/14



Sheet: /SHT4x/		
File: SHT4x.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 14/14